Geo-spatial Learning: How do we utilize multimodal data in the best way possible?

02501 Advanced Deep Learning in Computer Vision

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1 Introduction

Multimodal learning has become increasingly important in artificial intelligence, where models are trained on data originating from different sources to obtain richer and more robust representations. A well-known example is CLIP, which learns aligned representations of images and text using a contrastive learning objective. CLIP has since served as a powerful pretrained backbone for numerous downstream computer vision tasks. However, CLIP relies on only two modalities—images and text—while the real world contains far richer information, such as multispectral satellite imagery, GPS coordinates, audio, video, and more.

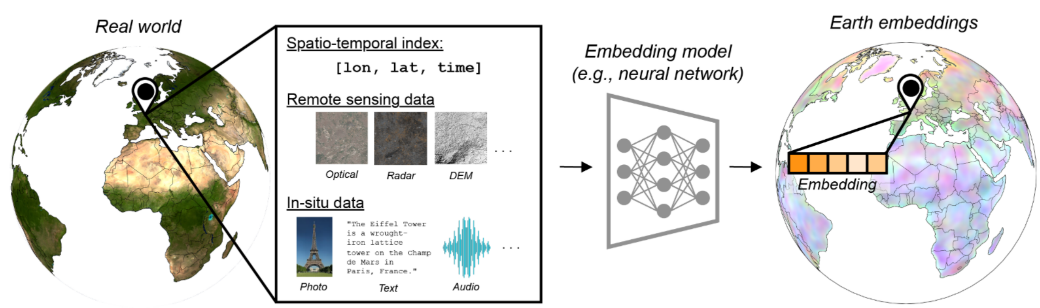

Figure 1: Multimodal data from various locations on earth offer great data complexity. The difficulty resides in training a model that effectively learns from all data sources. Taken from [1].

Recent efforts such as ImageBind [2] and LanguageBind [3] extend contrastive multimodal learning to a larger set of modalities by using a common "anchor" modality (typically images or text) to indirectly align all other data types. The resulting embedding space therefore only "indirectly" aligns the remaining modalities. We formulate a new research direction:

What if we investigate multimodal training strategies in **geo-spatial** tasks?

The geo-spatial field contains diverse data sources: ground-level photographs, aerial imagery, satellite observations, and geographic metadata—each offering complementary information that can improve performance and robustness. And there is a big push for creating reliable Earth Embeddings for representing our planet [1].

2 Tasks

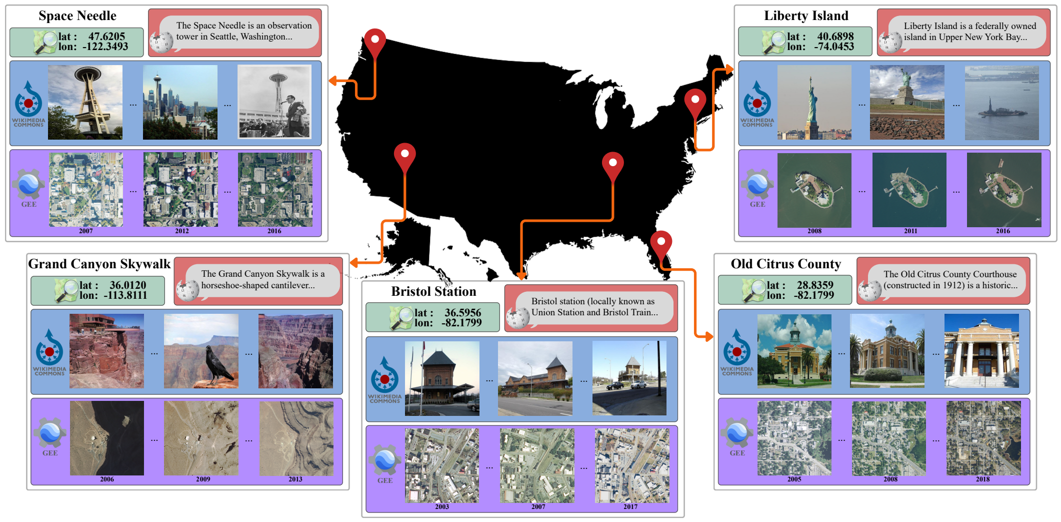

Figure 2: Multi-modal Landmarks dataset (MMLandmarks). Dataset collected from landmarks in the United States. Each landmark has multiple ground and aerial images, Wikipedia text, and GPS coordinates. Taken from [4].

In this project, we focus on geolocalization and/or cross-view retrieval, where the goal is to accurately locate images (Ground-to-GPS) or match them across different viewpoints (Ground-to-Satellite). Recent work suggests that incorporating aerial imagery substantially improves fine-grained localization, enabling models to predict locations within 100 meters [5]. Concurrently, the MMLandmarks [4] dataset is an open-source multimodal geo-spatial dataset, which contains ground-level images, aerial imagery, text information, and GPS coordinates collected across the United States. Using this dataset, investigate how additional modalities can be integrated into multimodal models to improve geolocalization and/or cross-view retrieval performance.

- Explore the data characteristics of MMLandmarks (geographical distribution, class imbalances, etc.)
- Implement a baseline geolocalization model (inspiration: StreetCLIP [6], GeoCLIP [7], OSV-5m [8] etc.).
- (and/or) Implement a simple cross-view localization baseline (inspiration: Sample4Geo [9], TransGeo [10], etc.).
- Extend the models by incorporating additional modalities:
  - Consider strategies for encoding Wikipedia text.
  - Investigate possible ways of encoding GPS information.
- Some additional elements to consider:
  - Do you want to use separate encoders?
  - What are the training objectives?
- Evaluate and compare the baselines against your multimodal models.
- How well do models adapt to other benchmarks (e.g. Uni-1652 [11] in cross-view, or make your own test set for geolocalization in the US)?

All the above are suggestions for directions of your project. You are most welcome to come with ideas that could tackle the problem from different angles.

References

[1] Konstantin Klemmer et al. “Earth Embeddings: Towards AI-centric Representations of our Planet”. In: (2025).

[2] Rohit Girdhar et al. “Imagebind: One embedding space to bind them all”. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2023, pp. 15180–15190.

[3] Bin Zhu et al. “Languagebind: Extending video-language pretraining to n-modality by language-based semantic alignment”. In: *arXiv preprint arXiv:2310.01852* (2023).

[4] Oskar Kristoffersen et al. “MMLANDMARKS: a Cross-View Instance-Level Benchmark for Geo-Spatial Understanding”. In: *arXiv preprint arXiv:2512.17492* (2025).

[5] Philipp Lindenberger et al. “Scaling Image Geo-Localization to Continent Level”. In: *NeurIPS*. 2025.

[6] Lukas Haas, Silas Alberti, and Michal Skreta. “Learning generalized zero-shot learners for open-domain image geolocalization”. In: *arXiv preprint arXiv:2302.00275* (2023).

[7] Vicente Vivanco Cepeda, Gaurav Kumar Nayak, and Mubarak Shah. “Geoclip: Clip-inspired alignment between locations and images for effective worldwide geo-localization”. In: *Advances in Neural Information Processing Systems* 36 (2023), pp. 8690–8701.

[8] Guillaume Astruc et al. “OpenStreetView-5M: The many roads to global visual geolocation”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2024, pp. 21967–21977.

[9] Fabian Deuser, Konrad Habel, and Norbert Oswald. “Sample4geo: Hard negative sampling for cross-view geo-localisation”. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 2023, pp. 16847–16856.

[10] Sijie Zhu, Mubarak Shah, and Chen Chen. “Transgeo: Transformer is all you need for cross-view image geo-localization”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2022, pp. 1162–1171.

[11] Zhedong Zheng, Yunchao Wei, and Yi Yang. “University-1652: A multi-view multi-source benchmark for drone-based geo-localization”. In: *Proceedings of the 28th ACM international conference on Multimedia*. 2020, pp. 1395–1403.